

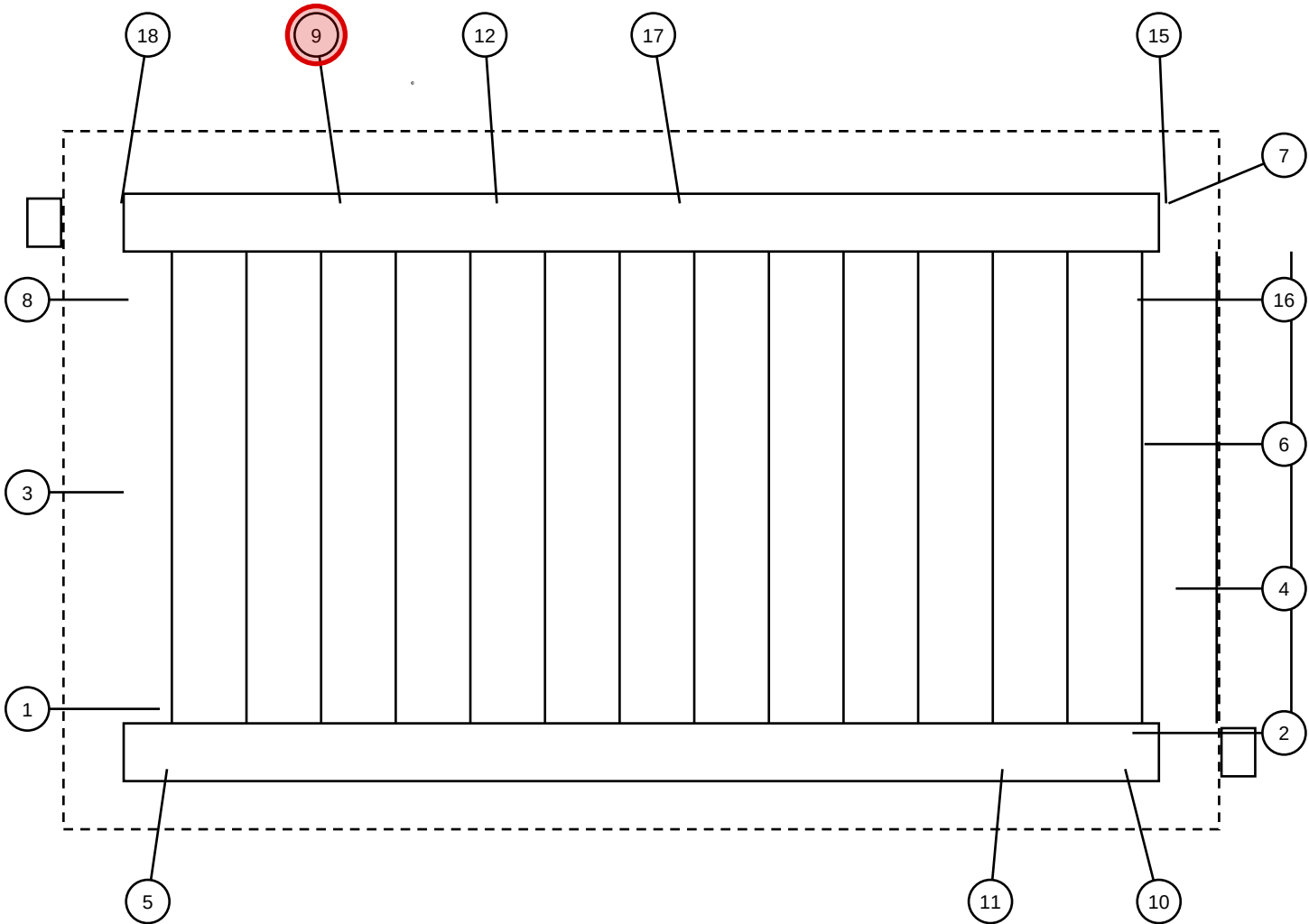
BOM CHECK (automated - verify before acting)

red = on drawing, missing in BOM

orange = qty / spec / revision mismatch

yellow = confirm match / extra / duplicate

blue = suggestion (implied item, rule-based)



MISSING IN BOM
DUPLICATE IN BOM

QTY: dwg 1 / BOM 2

Sr No	Part Description	Material	Size	Qty
1	INLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B		1
2	OUTLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B		1
3	ECONOMISER TUBE 38.1 OD X 3.6 THK	SA 210 GR.A1		96
4	RETURN BEND 38.1 OD	SA 210 GR.A1		42
5	TUBE SUPPORT PLATE 12 THK	IS 2062 GR.B		10
6	INLET FLANGE 200 NB CL.150	SA 105		1
7	OUTLET FLANGE 200 NB CL.150	SA 105		1
8	SAFETY VALVE 50 NB	SA 216 WCB		1
9	DRAIN VALVE 25 NB	SA 105		2
10	VENT VALVE 15 NB	SA 105		1
11	THERMOWELL 1/2 NPT	SS 316		2
12	PRESSURE TAPPING 1/2 NB	SA 105		2
13	GASKET 200 NB CL.150	SS 304 + GRAPHITE		2
14	STUD BOLT M20 X 90 WITH 2 NUTS	SA 193 B7		16
15	CASING PLATE 3 THK	IS 2062		20
16	EXPANSION JOINT 600 X 600	FABRIC		1
17	NAME PLATE	SS 304		1
18	INSPECTION DOOR 300 X 300	IS 2062		2

- NOTES:
- 1. ALL DIMENSIONS ARE IN MM.
 - 2. ALL WELDING AS PER IBR / ASME SEC I.
 - 3. PAINT: ONE COAT RED OXIDE PRIMER.

ECONOMISER ASSEMBLY

DRG NO : EC-13-GA-001
DRG REV : D
SCALE : NTS SIZE : A3
SAMPLE (SYNTHETIC) DRAWING

BOM CHECK SUMMARY

Drawing : drawing.pdf (Rev D)
BOM : bom.xlsx (Rev D)
Input : vector PDF
RUN RELIABILITY : HIGH

Drawing items checked : 18
BOM lines read : 21
BOM completeness : 94.4% (drawing items that have a BOM line)

Findings: duplicate 1, extra 1, omission 1, quantity 1

[RED] Item 9 'DRAIN VALVE 25 NB' (qty 2) is on the drawing but has no BOM line
[ORANGE] Item 16: quantity differs (drawing 1, BOM 2)
[YELLOW] BOM row 25 'Soot blower, retractable' has no matching item on the drawing (extra, or from another revision)
[YELLOW] BOM row 26 'Vent valve 15 NB' repeats item 10 (already covered by BOM row 14)

Automated check - not a substitute for engineering review. Blue items are rule-based suggestions.